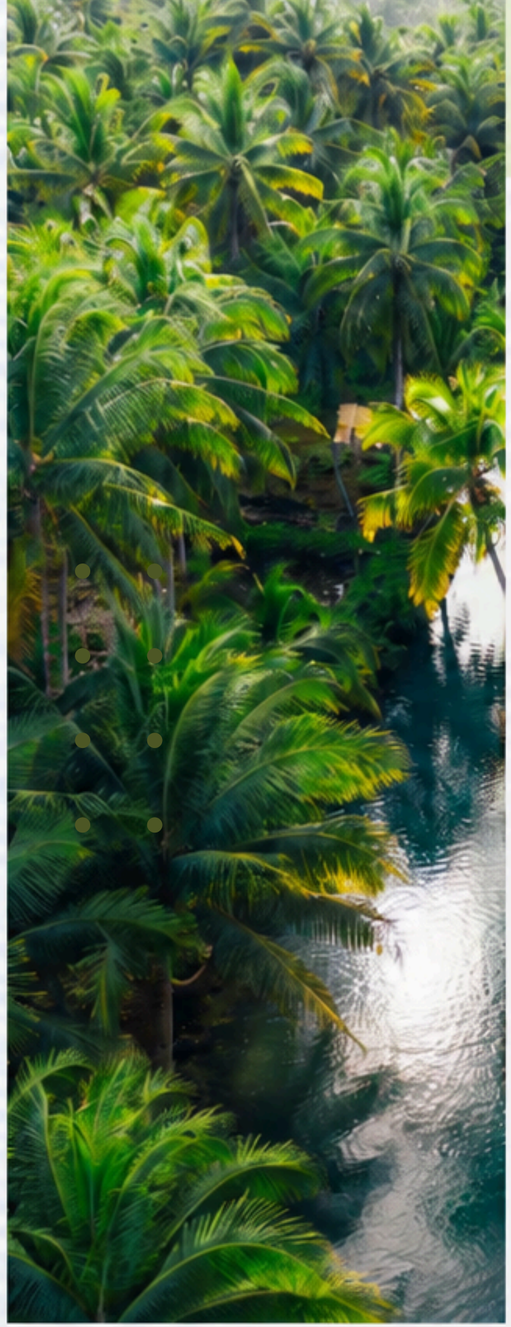
AI-Enabled Coconut Land Utilization, WCLWD Severity Assessment, Germplasm Identification and Community Intelligence Advisory System.

IT4010 Research Project – 2026

R26-IT-019

Comments for the Progress presentation -02

IT Number	Comment	Action
IT22904164	Improve the accuracy and validation performance of the component by evaluating multiple suitable models and comparing their results to identify the model that provides the best and most reliable performance.	An initial green masking algorithm method was used as a baseline, then refined using vegetation masking and spatial analysis. Comparing multiple detection models remains future work.
IT22236500	System should provide the expected yield production percentage based on the coconut tree type, with a clear explanation of how the prediction is calculated and validated.	My component identifies the coconut category from an image. Yield prediction requires additional information and actual yield records. These data were not available to me through CRI for this study, so I could not develop and validate a reliable yield estimate.
IT22549204	Further improve the accuracy and validation scores of the relevant component through better training data, parameter tuning, testing, and performance evaluation.	Fine-tuned the EfficientNet-B0 model and optimized hyperparameters to improve classification performance and evaluated the model using test metrics.
IT22088314	Method used to capture the affected area should be reconsidered to ensure accurate input. Poorly captured images may lead to incorrect affected-area estimation, so the system should include image-quality validation or a more reliable image-capturing mechanism before performing the analysis.	Implemented Mahalanobis-based Out-of-Distribution (OOD) detection to identify low-quality or unfamiliar input images and prevent unreliable disease severity assessments.



Problem & Solution

PROBLEM

Manual Identification
Coconut germplasm identification relies on visual observation and expert knowledge, which is time-consuming and prone to human error.

Disease Assessment
Identifying WCLWD symptoms and measuring disease severity accurately is challenging and usually requires expert field inspection.

Land Utilization
There is limited automated analysis of plantation occupancy, available land and potential planting capacity.

Fragmented Advisory
Disease information and management guidelines are scattered across different sources and languages, making it difficult for stakeholders to get timely, reliable advice.

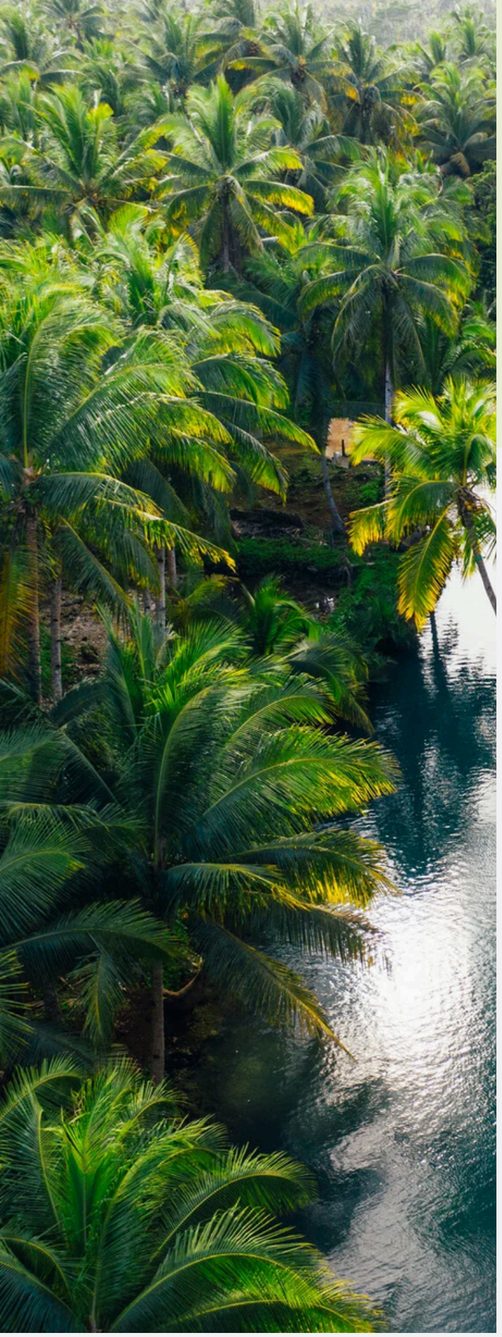
SOLUTION

AI-Based Germplasm Identification
Classifies Dwarf, King Coconut and Tall varieties using deep learning with visual explanations (Grad-CAM) to support germplasm identification.

Automated WCLWD Severity Assessment
Uses AI classification, weakly supervised segmentation and ABCD-inspired features to compute a Disease Severity Index and classify severity levels.

Smart Land Utilization Analysis
Detects coconut palms from aerial images, analyses land utilization and estimates approximate planting capacity for better plantation planning.

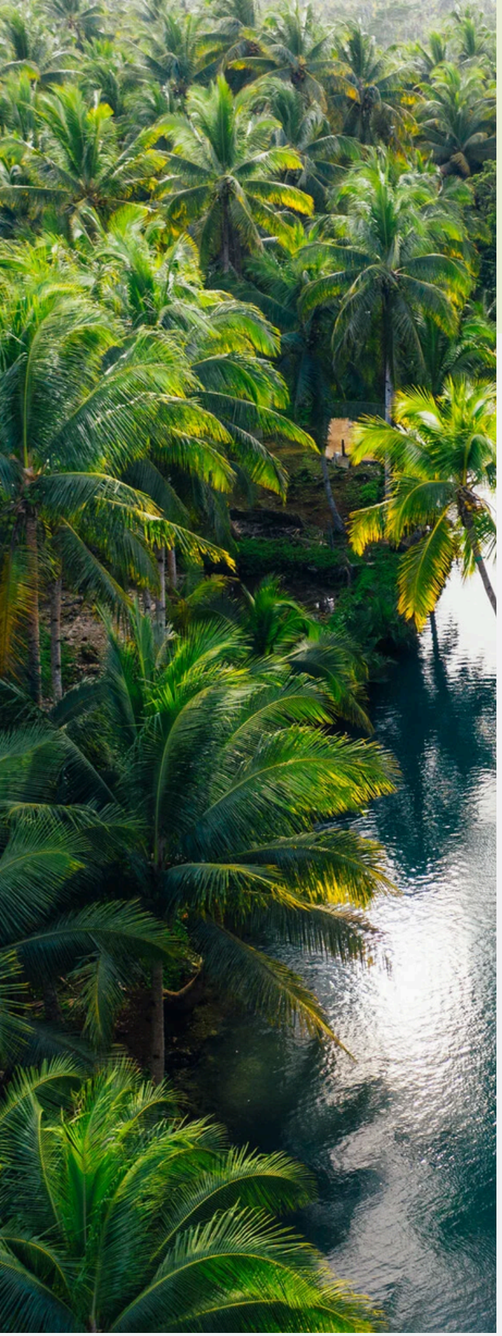
Intelligent Disease Advisory
Classifies coconut leaf diseases and provides retrieval-based treatment and management guidance in English and Sinhala through an AI-powered chatbot.



IT22904164 - Wijerathna W.M.H.N.

Land Utilization Analysis System for Coconut Genebanks

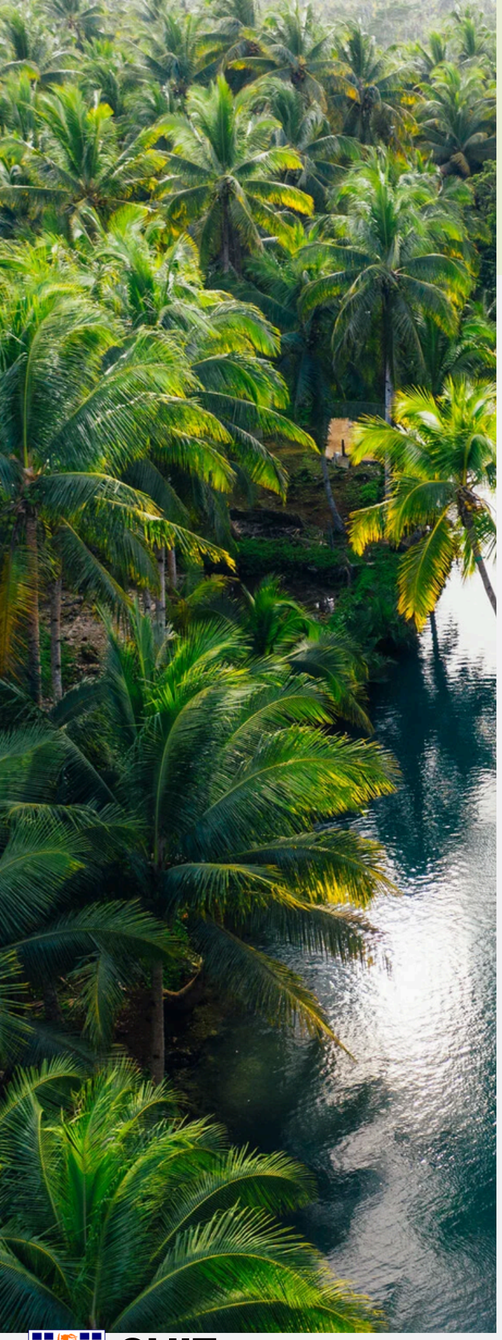
This component analyzes aerial images to detect palms/trees and evaluate land utilization. It uses YOLO11n for detection, Excess Green masking to estimate occupied and available land, distance-based analysis to find free spaces, and a greedy algorithm to estimate additional planting positions.



IT22236500 - Randeniya A.A.S.L.B.R.P.W.R.C.

AI-Based Image Classification System for Coconut Germplasm Identification

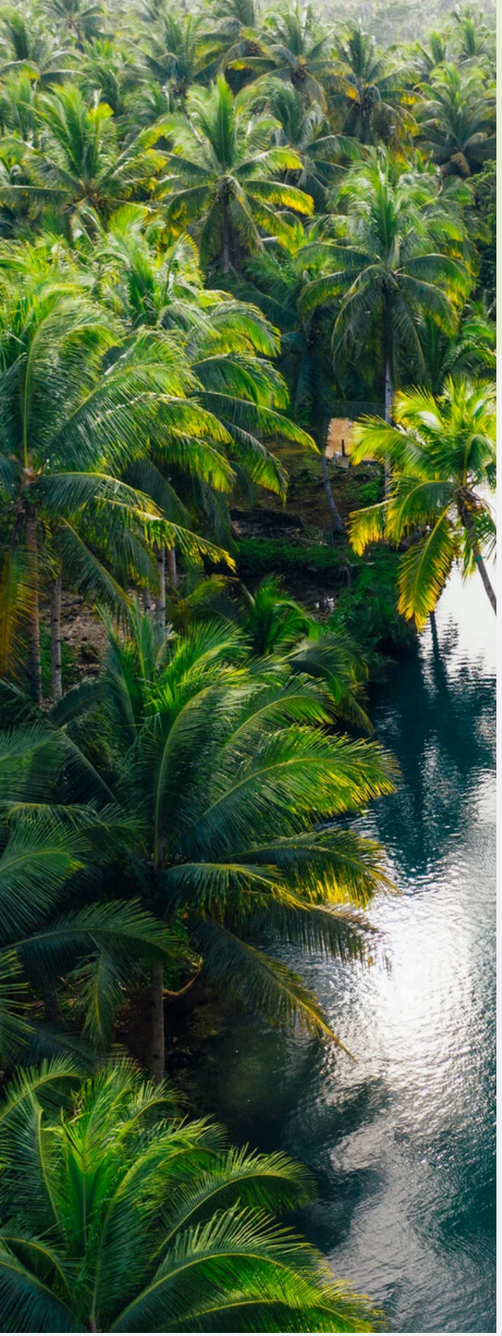
This component analyzes coconut images to classify them as Dwarf, King Coconut, or Tall. It uses an ImageNet-pretrained EfficientNet-B0 to extract visual features, prototype-based classification with cosine similarity to predict the category, and Grad-CAM to highlight image regions associated with the prediction.



IT22549204 - Harischandra R.D.I.D.

AI-Based Coconut Disease Identification and Intelligent Advisory System

This component integrates AI-based coconut leaf disease classification using EfficientNet-B0 with a RAG-powered intelligent advisory chatbot. It identifies four leaf conditions, detects unfamiliar images, and provides knowledge-based management guidance in English and Sinhala.



IT22088314 - Malshani W.A.M.S.

Weligama Coconut Leaf Wilt Disease Severity Assessment Using ABCD Rule

This component automates Weligama Coconut Leaf Wilt Disease (WCLWD) severity assessment using Weakly Supervised Semantic Segmentation (WSSS) and the ABCD rule to analyze infected leaf regions, extract disease features, calculate a Disease Severity Index (DSI), and classify disease severity.

Results and Testimonials

System Performance Overview

Evaluation metrics and dataset splits

C1 — Germplasm

96.53%

Episodiic validation accuracy

50 episodes drawn from an 87-image validation partition of a 428-image dataset

C2 — WCLWD severity

98.79% (0.9877)

Test accuracy (weighted F1)

497-image group-separated test partition of a 3,354-image dataset

C3 — Aerial detector

83.5%

mAP@50

42-image validation partition of a 279-image, 10,398-box dataset

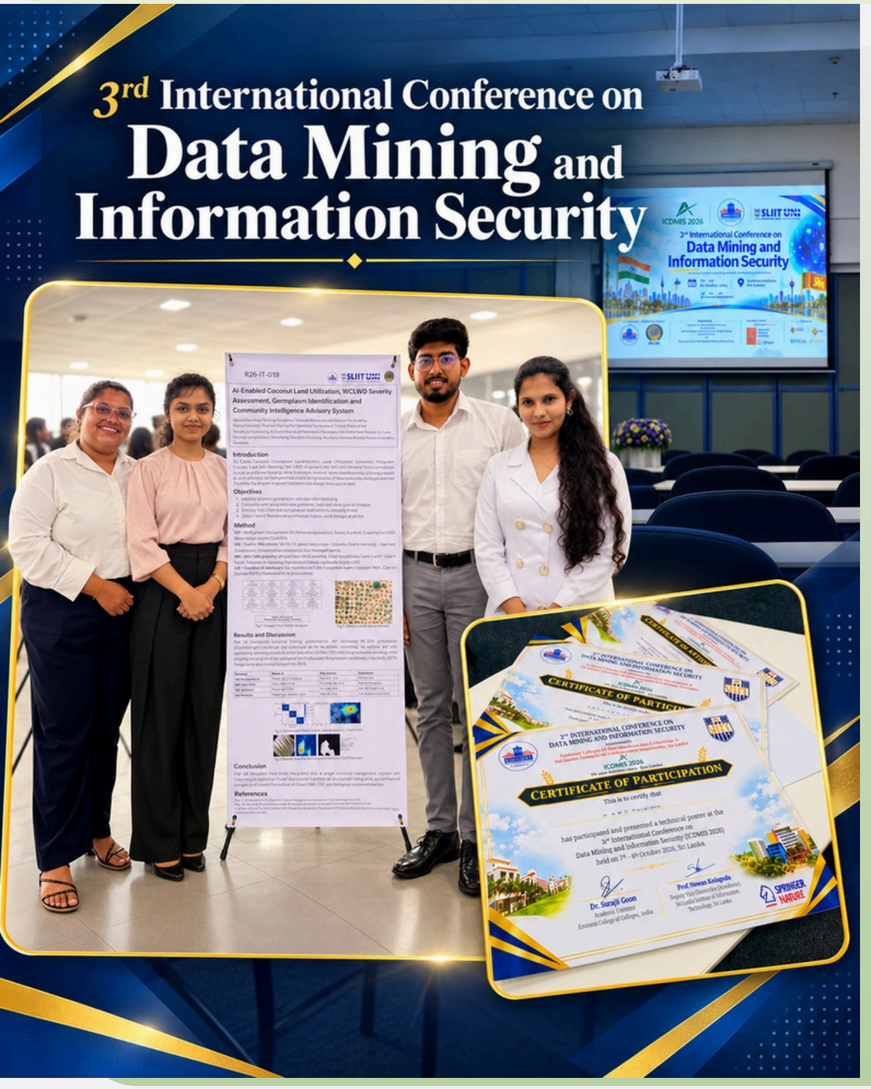
C4 — Disease classifier

98.89%

Test accuracy

723-image test partition of 4,807 images

Achievements



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THANK YOU!